



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER II, SESSION 2024/2025

BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)

SECB2103 BIOINFORMATICS I - SECTION 01

PROJECT

FIND A GENE

GROUP NAME: DATA VOYAGERS

GROUP MEMBERS	MATRIC NO
CHIN PEI WEN	A23CS0065
KOO XUAN	A23CS0300
TAN ZHAO HONG	A23CS0188

LECTURER'S NAME : DR HASLINA BINTI HASHIM

SUBMISSION DATE : 03 JULY 2025

PRESENTATION VIDEO LINK: <https://youtu.be/8OfMDVVGxJQ>

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STEP 1 Protein of interest

Tell me the name of a protein you are interested in.

Name of protein: Ferritin heavy chain

Species: Homo sapiens

Accession number: NP_002023.2

STEP 2 BLAST search using TBLASTN

The screenshot shows the NCBI TBLASTN search interface. The top navigation bar includes links for blastn, blastp, blastx, **tblastn**, and tblastx. The main heading is "TBLASTN search translated nucleotide databases using a protein query, more...". There are "Reset page" and "Bookmark" buttons in the top right.

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

Query subrange [?](#)

From

To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ?

Organism ? ☐ exclude [Add Organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to ☐ Sequences from type material

Entrez Query [View Help](#) [Create custom database](#)

Enter an Entrez query to limit search [?](#)

BLAST ☐ Search database est using Tblastn (search translated nucleotide databases using a protein query)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign

[+ Algorithm parameters](#)

[Feedback](#)

Figure 2.1 TBLASTN page

Paste the accession number of Ferritin protein into the box

i Your search is limited to records that include: Nematoda (taxid:6231)

Job Title	NP_002023:ferritin heavy chain [Homo sapiens]
RID	4ZPYVH9Z016 Search expires on 06-17 16:54 pm Download All ▼
Program	TBLASTN ? Citation ▼
Database	est See details ▼
Query ID	NP_002023.2
Description	ferritin heavy chain [Homo sapiens]
Molecule type	amino acid
Query Length	183
Other reports	?

Filter Results

Organism only top 20 will appear ☐ exclude

[+ Add organism](#)

Percent Identity to
E value to
Query Coverage to

[Filter](#) [Reset](#)

Descriptions	Graphic Summary	Alignments	Taxonomy
---------------------	-----------------	------------	----------

Sequences producing significant alignments [Download](#) ▼ [Select columns](#) ▼ Show [?](#)

☒ select all 100 sequences selected [GenBank](#) [Graphics](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	BSBmMFSZ16B059K Brugia malayi microfilaria cDNA (SAW94LS-BmMf) Brugia malayi cDNA clone BSBmMFSZ...	Brugia malayi	331	331	90%	3e-115	95.73%	775	AI939639.1
☒	Ac_EH1f_01C10_M13 Adult Anguillicola crassus Anguillicola crassus cDNA clone Ac_EH1f_01C10_mRNA seque...	Anguillicola crass...	260	260	94%	2e-87	68.60%	747	JK511428.1
☒	kc93f10.y2 Xiphinema index CSEQDL01 Xiphinema index cDNA 5' similar to SW:FRIS_LYMST_P42577 SOMAFE...	Xiphinema index	224	224	96%	7e-74	60.23%	633	CV579891.1
☒	kc52f03.y1 Xiphinema index CSEQDA01 Xiphinema index cDNA 5' similar to SW:FRIS_LYMST_P42577 SOMAFE...	Xiphinema index	224	224	96%	9e-74	60.23%	633	CV509916.1
☒	AIAC-aaa91d02.b1 Ancylostoma caninum EST Male pSMART Ancylostoma caninum cDNA similar to sp P4257...	Ancylostoma can...	223	223	91%	6e-73	62.28%	685	EX539578.1
☒	C5_TOMix_AFG04_TW C5_TOMix Caenorhabditis sinica cDNA clone C5_TOMix_AFG04 similar to UniRef100_Q...	Caenorhabditis si...	217	217	94%	6e-71	61.05%	625	FD514558.1
☒	C5_TOMix_R0H11_TW C5_TOMix Caenorhabditis sinica cDNA clone C5_TOMix_R0H11 similar to UniRef100_Q...	Caenorhabditis si...	216	216	94%	7e-71	61.05%	630	FD517063.1
☒	C5_TOMix_AFG01_TW C5_TOMix Caenorhabditis sinica cDNA clone C5_TOMix_AFG01 similar to UniRef100_Q...	Caenorhabditis si...	216	216	94%	9e-71	61.05%	639	FD514555.1
☒	kc56a12.y1 Xiphinema index CSEQDA01 Xiphinema index cDNA 5' similar to TR:O61915_O61915 FERRITIN_mr...	Xiphinema index	217	217	89%	1e-70	61.96%	705	CV510188.1
☒	C5_TOMix_K0B04_TW C5_TOMix Caenorhabditis sinica cDNA clone C5_TOMix_K0B04 similar to UniRef100_Q9...	Caenorhabditis si...	216	216	94%	1e-70	62.21%	648	FD516347.1
☒	C5_TOMix_AFG03_TW C5_TOMix Caenorhabditis sinica cDNA clone C5_TOMix_AFG03 similar to UniRef100_Q...	Caenorhabditis si...	216	216	94%	1e-70	61.05%	645	FD514557.1
☒	CPAD-aab03h02.b1 PB2801_EST_CPAD1 Caenorhabditis brenneri cDNA similar to ref NP_491198.1 Ferritin (19...	Caenorhabditis b...	216	216	94%	1e-70	60.47%	641	FF090610.1

Figure 2.2 Result of blast search

This shows the result of the TBLASTN for ferritin protein



Figure 2.3 Graphic summary of the ferritin protein TBLASTN search

[Download](#) [GenBank](#) [Graphics](#) [Next](#) [Previous](#) [Descriptions](#)

Ac_EH1f_01C10_M13 Adult Anguillicola crassus Anguillicola crassus cDNA clone Ac_EH1f_01C10, mRNA sequence

Sequence ID: [JK511428.1](#) Length: 747 Number of Matches: 1

Range 1: 135 to 650 [GenBank](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps	Frame
260 bits(664)	2e-87	Compositional matrix adjust.	118/172(69%)	147/172(85%)	0/172(0%)	+3
Query 6	TSQVRQNYHQDSEAAINRQINLELYASYVYLSMSYFDRDDVALKNFAKYFLHQSHREERE	65				
Sbjct 135	TSQVRQNYH+D EAAINR N+EL+ASY Y+S+H++YF RDDVAL F+ +F S EERE	314				
Query 66	HAELMKLQNRGGRIFLQDIKKPDCDDWESGLNAMECALHLEKNVQSLLELHKLATDK	125				
Sbjct 315	HAELK+ QN+RGGRIFL D+KKP+ D+W SGL AM+CAL LEK+VNQ+LL+LHK+A+DK	494				
Query 126	NDPHLCDFIETHYLNQVKAIKELGDHVTNLRKMGAPESGLAEYLFDKHTLG	177				
Sbjct 495	DPHLCDF+ETHYLNQV+AIK+LGD++TNL +M + + +AEYLFDKHTLG	650				

Figure 2.4 Query of ferritin heavy chain protein

Query: ferritin heavy chain [Homo sapiens] Query ID: NP_002023.2 Length: 183

>Ac_EH1f_01C10_M13 Adult Anguillicola crassus Anguillicola crassus cDNA clone
Ac_EH1f_01C10, mRNA sequence

Sequence ID: JK511428.1 Length: 747

Range 1: 135 to 650

Score:260 bits(664), Expect:2e-87,

Method:Compositional matrix adjust.,

Identities:118/172(69%), Positives:147/172(85%), Gaps:0/172(0%)

Query	6	TSQVRQNYHQDSEAAINRQINLELYASYVYLSMSYYFDRDDVALKNFAKYFLHQSHEERE	65
		TSQ+RQNYH+D EAAINR N+EL+ASY Y+SM++YF RDDVAL F+ +F S EERE	
Sbjct	135	TSQIRQNYHRDCEAAINRMANMELFASYTYMSMAHYFARDDVALPGFSHFFKENSEEERE	314
Query	66	HAEKLMKLQNQRGGRIFLQDIKKPDCDDWESGLNAMECALHLEKNVNQSLLELHKLATDK	125
		HAEKL+ QN+RGGRIFL D+KKP+ D+W SGL AM+CAL LEK+VNQ+LL+LHK+A+DK	
Sbjct	315	HAEKLLSFQNKRGGRIFLNDVKKPERDEWGSGLAMQCALQLEKDVNQALLDLHKIASDK	494
Query	126	NDPHLCDFIETHYLNEQVKAIKELGDHVTNLRKMGAPESGLAEYLFDKHTLG	177
		DPHLCDF+ETHYLNEQV+AIK+LGD++TNL +M + + +AEYLFDKHTLG	
Sbjct	495	IDPHLCDFLETHYLNEQVEAIKKLGDIYITNLARMDSGNNKMAEYLFDKHTLG	650

STEP 3 Sequence of novel protein

```
>Anguillicola_crassus_ferritin (sequence taken from BLAST result)
TSQIRQNYHRDCEAAINRMANMELFASYTYMSMAHYFARDDVALPGFSHFFKENSEEEREHAEKLLSFQNKRGGRIFL
NDVKKPERDEWGSGLAMQCALQLEKDVNQALLDLHKIASDKIDPHLCDFLETHYLNEQVEAIKKLGDIITNLARMDS
GNNKMAEYLFDKHTLG
```

Name: Ferritin-like protein

Organism: *Anguillicola crassus*

Species:

Eukaryota; Metazoa; Ecdysozoa; Nematoda; Chromadorea; Spirurida; Dracunculoidea;
Anguillicolidae; *Anguillicola*.

STEP 4 Confirm novel protein via BLASTP

-If there is a match with 100% amino acid identity to a protein in the database, from the same species, then the protein is NOT novel even if the match is to a protein with a name such as “unknown”. Someone has already found and annotated this sequence, and assigned it an accession number.

-If there is a match with less than 100% identity, then it is likely that your protein is novel, and you have succeeded.

-If there is a match with 100% identity, but to a different species than the one you started with, then you have succeeded in finding a novel gene.

-If there are no database matches to the original query from step[1], this indicates that you have partially succeeded: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

blastn **blastp** blastx tblastn tblastx

BLASTP programs search protein databases using a protein query. more...

Reset page Bookmark

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ? Clear

>Anguillicola_crassus_ferritin
TSQIRQNYHRDCEAAINRMANMELFASYTMSAHYFARDVDVALPGFSHF
KENSEEERE
HAEKLLSFQNKRGGRIFLNDVKKPERDEWGSQLEAMQCALQLEKDVNQALL

Query subrange ? From To

Or, upload file Choose File No file chosen ?

Job Title Anguillicola_crassus_ferritin

Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database ☒ Standard databases (nr etc.):
☐ ClusteredNR database **RECOMMENDED** Learn more...

Non-redundant protein sequences (nr) ?

Organism Optional Nematoda (taxid:6231) exclude Add organism

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown ?

Exclude Optional ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm ☐ Quick BLASTP (Accelerated protein-protein BLAST)
☒ blastp (protein-protein BLAST)
☐ PSI-BLAST (Position-Specific Iterated BLAST)
☐ PHI-BLAST (Pattern Hit Initiated BLAST)
☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm ?

Feedback

Figure 4.1 BLASTP page

National Library of Medicine
National Center for Biotechnology Information

Log in

BLAST® » blastp suite » results for RID-4ZRC1R81013

HomeRecent ResultsSaved StrategiesHelp

come the default Protein BLAST database. [Learn more about ClusteredNR](#)

Edit Search
Save Search
Search Summary

How to read this report?
BLAST Help Videos
Back to Traditional Results Page

! Your search is limited to records that include: Nematoda (taxid:6231)

Job TitleAnguillicola_crassus_ferritin
RID4ZRC1R81013 Search expires on 06-17 17:01 pm Download All
ProgramBLASTP Citation
Databasenr See details
Query IDlcl|Query_7692989
DescriptionAnguillicola_crassus_ferritin
Molecule typeamino acid
Query Length172
Other reportsDistance tree of resultsMultiple alignmentMSA viewer

Filter Results

Organism only top 20 will appear
☐ exclude

Type common name, binomial, taxid or group name

Add organism

Percent Identity

E value

Query Coverage

to

to

to

Filter
Reset

Compare these results against the new Clustered nr database
BLAST

Descriptions
Graphic Summary
Alignments
Taxonomy

Sequences producing significant alignments
Download
Select columns
Show 100

select all 100 sequences selected
GenPept
Graphics
Distance tree of results
Multiple alignment
MSA Viewer

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Soma ferritin [Trichinella pseudospiralis]	Trichinella pseudospiralis	217	217	99%	2e-71	60.23%	172	KAL1231357.1
☒	Soma ferritin [Trichinella spiralis]	Trichinella spiralis	217	217	99%	2e-71	60.23%	172	KAL1242961.1
☒	Soma ferritin [Trichinella pseudospiralis]	Trichinella pseudospiralis	216	216	99%	8e-71	59.65%	172	KRX93966.1
☒	Soma ferritin [Trichinella nativa]	Trichinella nativa	215	215	99%	2e-70	59.06%	172	KRZ57361.1
☒	Soma ferritin [Trichinella nelsoni]	Trichinella nelsoni	212	212	99%	3e-69	59.20%	175	KRX18037.1
☒	Soma ferritin, partial [Trichinella zimbabwensis]	Trichinella zimbabwensis	212	212	99%	3e-69	58.62%	175	KRZ14145.1

Figure 4.2 Result of BLASTP for *Anguillicola crassus* ferritin

Descriptions

Graphic Summary

Alignments

Taxonomy

Alignment view

Pairwise

Restore defaults

Download

100 sequences selected

Download

GenPept

Graphics

Next

Previous

Descriptions

Soma ferritin [Trichinella pseudospiralis]

Sequence ID: [KAL1231357.1](#) Length: 172 Number of Matches: 1

Range 1: 2 to 169

GenPept

Graphics

Next Match

Previous Match

Score	Expect	Method	Identities	Positives	Gaps
217 bits(553)	2e-71	Compositional matrix adjust.	103/171(60%)	128/171(74%)	3/171(1%)
Query 2	SQIRQNYHRDCEAAINRMANMELFASYTYMSAHYFARDQVALPGFSHFFKENSEEEREH	61			
Sbjct 2	S I QNH + EAAH+ N+EL+ASY Y+SMA YF RDDVALPGFS FF + SEER H	61			
Query 62	AEKLLSFQNKRGGRIFLNDVKKPERDEWGSGLAEMQCALQLEKDVNQALLDLHKIASDKI	121			
Sbjct 62	A K + +Q RGR+ ++ KP DEWGSGLAEM AL LEK VNQ+LLDLH IAS	121			
Query 122	DPHLCDLFETHYLNQVEAIKKLGDYITNLARMDSGNKMAEYLFQKHTLG	172			
Sbjct 122	D +L DFLE +L EQ+E+IK+L DY+TNL R+ +G + EYLFQ+HTLG	169			

Download

GenPept

Graphics

Next

Previous

Descriptions

Soma ferritin [Trichinella spiralis]

Sequence ID: [KAL1242961.1](#) Length: 172 Number of Matches: 1

Range 1: 2 to 169

GenPept

Graphics

Next Match

Previous Match

Score	Expect	Method	Identities	Positives	Gaps
217 bits(553)	2e-71	Compositional matrix adjust.	103/171(60%)	128/171(74%)	3/171(1%)
Query 2	SQIRQNYHRDCEAAINRMANMELFASYTYMSAHYFARDQVALPGFSHFFKENSEEEREH	61			
Sbjct 2	S I QNH + EAAH+ N+EL+ASY Y+SMA YF RDDVALPGFS FF + SEER H	61			
Query 62	AEKLLSFQNKRGGRIFLNDVKKPERDEWGSGLAEMQCALQLEKDVNQALLDLHKIASDKI	121			
Sbjct 62	A K + +Q RGR+ ++ KP DEWGSGLAEM AL LEK VNQ+LLDLH IAS	121			
Query 122	DPHLCDLFETHYLNQVEAIKKLGDYITNLARMDSGNKMAEYLFQKHTLG	172			
Sbjct 122	D +L DFLE +L EQ+E+IK+L DY+TNL R+ +G + EYLFQ+HTLG	169			

Figure 4.3 *Anguillicola crassus* is novel by comparing it against the non-redundant (nr) protein database using BLASTP.

The BLASTP results show that the *Anguillicola crassus* ferritin-like protein has only 60% identity to ferritin proteins from other nematode species which are *Trichinella pseudospiralis* and *T. spiralis*, and there's no exact matches were found in the database. Therefore, this protein is considered novel for the purposes of this study, as it is sufficiently different from known proteins in public databases.

STEP 5 Multiple sequence alignment

Our original query protein:

```
>NP_002023.2 ferritin heavy chain [Homo sapiens]
MTTASTSQVRQNYHQDSEAAINRQINLELYASYVYLSMSYYFDRDDVALKNFAKYFLHQSHEEREHAEKL
MKLQNRQGGRIFLQDIKKPDCDDWESGLNAMECALHLEKNVNQSLLELHKLATDKNDPHLCDFIETHYLN
EQVKAIKELGDHVTNLRKMGAPESGLAEYLFDKHTLGSDSDNES
```

Our novel protein:

```
>Anguillicola_crassus_ferritin (sequence taken from BLAST
result)
TSQIRQNYHRDCEAAINRMANMELFASYTYMSMAHYFARDDVALPGFSHFFKENSEEEREHAEK
LLSFQNKRGGRIFLNDVKKPERDEWGSGLAMQCALQLEKDVNQALLDLHKIASDKIDPHLCDF
LETHYLNQVEAIKKLGDIYITNLARMDSGNNKMAEYLFDKHTLG
```

Below is the list of 7 protein sequences selected by our group for the Multiple Sequence Alignment :

```
>NP_001084057.1 ferritin heavy chain A [Xenopus laevis]
MQSQVRQNFHSDCEAAINRMVNMEMYASYVYLSMSYYFDRDDVALHHVAKFFKEQSHEEREHAEKFLKYQNKRGGRVV
LQDIKKPERDEWGNLTLEATQAALQLEKTVNQALLDLHKLASDKVDAHLCDFLESEYLEEQVKAMQLGDYITNLKRLG
VPQNGMGEYLFDKHTLGESS
```

```
>NP_001316112.1 ferritin heavy chain-like precursor [Diaphorina citri]
MNSVLLNLSV FALLSIVSVSQAAQDSTCSAFVNQKVCSKKDQSVEIDKLLGVSCSAQYEGFSAIQPKIRQYINAQIID
SYQLTLLSTNFGNHKANRPGFEALYKKLSDEAWEDAIELIKYNAKRGGLVDFSDIRGRDLKSMVDEQKKS AITEYNS
LATAVDLQKELAAGAHDIHKEAIRLGQAYHDPEISSFIEKEFVHKYSSNIRTLAGYLADMNSIMSEYTENGGPSLSLY
LFDEY LQKSLGVAAA
```

```
>XP_015719168.1 ferritin heavy chain [Coturnix japonica]
MATPPSQVRQNYHQDCEAAINRQINLELYASYVYLSMSYYFDRDDVALKNFAKYFLHQSHEEREHAEKLMKLQNRGG
RIFLQDIKKPDRDDWENGLTAMECALHLEKNVNQSLLELHKLATEKNDPHLCDFIETHYLDEQVKAIKQLGDHVTNLR
KMGAPKYGMAEYLFDKHTLGECDNS
```

```
>XP_023012497.2 ferritin heavy chain [Leptinotarsa decemlineata]
MKSIIILTTFFICVIGLASSALECSYKELDIPKDWIDMEKACVKKMRAQVEDELKAAMQYMAMGAHF SKD TVNRP GF AE
IFFKSASEEREHAIKLISYLLMRGELTSKVSSLIKRNLMPSQTTWTNGLSALKDALKLEASVTRKIRDVIKVC EEAKS
FNDYHLVDYLSGDFLGEQYQGQRDIAGKISTLEKMTEKHGALGEWLFDKLLKGEL
```

```
>XP_040229091.2 ferritin subunit [Anopheles coluzzii]
MMKSIFFGIVALMFAAVVMQQDQASAQVTETDAPSSTDEWNYMNRSCSAKLQDQINKEFDAAIFYMQYGAYFAQYQVN
LPGFEKFFFNAASEEREHGMKLI EYALMRGQKPIDRNSFSLNFANPAARVDAEQGSVALTALK AALAKEQEVTKSIRE
LIKICEEDHNDYHLVDYLTGEFLEE QHQGQRDLAGKITMLSKLLRTNPKLGEFMFDKQNM
```

>XP_045593834.2 LOW QUALITY PROTEIN: ferritin heavy chain [Procambarus clarkii]

MALVIQXWLLVIGIIAVQATTGSDLEYNCMPDNGDKVPKEVCYIQHCAKAIEGQIKEELNAAFMYMGIRFGQYTV
DRPGITKFLMEAATEERSHAILMLDYLNTRGINLSTLSYSFQFKNYTVYLNMMYRDALKEALNMEIEVTKLIYDVV
SACGNDYHGADVFTNPILDEQHKGVKRLQALRAFDDLSENYTGNAVLAEYIFDQKMLSGEFLS

>XP_049280091.1 ferritin subunit [Anopheles funestus]

MMKSIFFGIVALMFAAVVMQQEQASAVTDTQPSSTDEWNYMNRSCSAKLQDQINKEFDAAIFYMQYAAQYKVN
LPGFENFFFKAASEEREHGMKLI EYALMRGQKPIDRNTFSLHFANQVQQPEPGPEGSVALTALKGALAKEQE

Simplify original query protein, novel protein and 7 proteins of other family:

>Homo sapiens NP_002023.2 ferritin heavy chain

MTTASTSQVRQNYHQDSEAAINRQINLELYASYVYLSMSYYFDRDDVALKNFAKYFLHQSHEEREHAEKLMKLQNQRG
GRIFLQDIKKPDCDDWESGLNAMECALHLEKNVNQSLLELHKLATDKNDPHLCDFIETHYLNEQVKA I KELGDHVTNL
RKMGAPESGLAEYLFDKHTLGDSDNES

>(Novel)Anguillicola_crassus_ferritin (sequence taken from BLAST result)

TSQIRQNYHRDCEAAINRMANMELFASYTYMSMAHYFARDDVALPGFSHFKEENSEEREHAEKLLSFQNKRGGRIFL
NDVKKPERDEWGSGLAMQCALQLEKDVNQALLDLHKIASDKIDPHLCDFLETHYLNEQVEAIKKLGDYITNLARMDS
GNNKMAEYLFDKHTLG

>Xenopus laevis NP_001084057.1 ferritin heavy chain A

MQSQVRQNFHSDCEAAINRMVNMEMYASYVYLSMSYYFDRDDVALHHVAKFFKEQSHEEREHAEKFLKYQNKRGRV
LQDIKKPERDEWGNLTLEATQAALQLEKTVNQALLDLHKLASDKVDAHLCDFLESEYLEEQVKAMKQLGDYITNLKRLG
VPQNGMGEYLFDKHTLGESS

>Diaphorina citri NP_001316112.1 ferritin heavy chain-like precursor

MNSVLLNLSV FALLSIVSVSQAQDSTCSAFVNQKVCSSKQDQSV EIDKLLGVSCSAQYEGFSAIQPKIRQYINAQIID
SYQLTLLSTNFGNHKANRPGFEALYKKLSDEAWEDAIELIKYNAKRGKLVDFSDIRGRDLKSMVDEQKKS AITEYNS
LATAVDLQKELAAAGHDHKEAIRLGQAYHDPEISSFIEKEFVHKYSSNIRTLAGYLADMNSIMSEYTENGGPSLSLY
LFDEYLQKSLGVAAA

>Coturnix japonica XP_015719168.1 ferritin heavy chain

MATPPSQVRQNYHQDCEAAINRQINLELYASYVYLSMSYYFDRDDVALKNFAKYFLHQSHEEREHAEKLMKLQNQRGG
RIFLQDIKKPDRDDWENGLTAMECALHLEKNVNQSLLELHKLATEKNDPHLCDFIETHYLDEQVKA I KQLGDHVTNLR
KMGAPKYGMAEYLFDKHTLGECDNS

>Leptinotarsa decemlineata XP_023012497.2 ferritin heavy chain

MKSIILTTFFICVIGLASSALECSYKELDIPKDWIDMEKACVKKMRAQVEDELKAAMQYMAMGAHFSKDTVNRPGFAE
IFFKSASEEREHAIKLISYLLMRGELTSKVSSLIKRNLMPSQTTWTNGLSALKDALKLEASVTRKIRDVIKVCEEAKS
FNDYHLVDYLSGDFLGEQYQGQRDIAGKISTLEKMTEKHGALGEWLFDKKLLKGEL

>Anopheles coluzzii XP_040229091.2 ferritin subunit

MMKSIFFGIVALMFAAVVMQQDQASAVTETDAPSSTDEWNYMNRSCSAKLQDQINKEFDAAIFYMQYGAYFAQYQVN
LPGFEKFFFNAASEEREHGMKLI EYALMRGQKPIDRNSFSLNFANPAARVDAEQGSVALTALKAAALAKEQEVTKSIRE
LIKICEEDHNDYHLVDYLTGEFLEEQHQQRDLGKITMLSKLLRTNPKLGEFMFDKQNM

>Procamburus clarkii XP_045593834.2 LOW QUALITY PROTEIN: ferritin heavy chain
MALVIQXWLLVIGIIAVQATTGSDLEYNCMPDNGDKVPKEVCYIQHCAKAIEGQIKEELNAAFMYMGIRFGQYTV
DRPGITKFLMEAATEERSHAILMLDYLNTRGINLSTSLSYSFQFKNYTVYLNMMYRDALKEALNMEIEVTKLIYDVV
SACGNDYHGADVFTNPILDEQHKGVKRLQALRAFDDLSENYTGNAVLAEYIFDQKMLSGEFLS

>Anopheles funestus XP_049280091.1 ferritin subunit
MMKSIFFGIVALMFAAVVMQQEQASAVTDTQPSSTDEWNYMNRSCSAKLQDQINKEFDAAIFYMQYAAFYAQYKVN
LPGFENFFFKAASEEREHGMKLI EYALMRGQKPIDRNTFSLHFANQVQQPEPGPEGSVALTALKGALAKEQE

Result using MAFFT at EBI in Clustal form:

>NP_002023.2 ferritin heavy chain [Homo sapiens]
MTTA-----STSQVRQNYHQ-----DSEA---
-----AINRQINLELYASYVYLSMSYYFDRDDVALKNFAKYFLHQ SHEEREHA EKLMK
LQNQRGGRIFL-QDIKK-----PDCDDWESG---LNAMECALHLEKNVNQSLLELHKL
ATDK----NDPHLCDFIETHYLNEQVKAIKELGDHVTNLRKMGAPESG-----LAEYLF D
---KHTL-GDSDNES
>XP_015719168.1 ferritin heavy chain [Coturnix japonica]
MAT-----PPSQVRQNYHQ-----DCEA---
-----AINRQINLELYASYVYLSMSYYFDRDDVALKNFAKYFLHQ SHEEREHA EKLMK
LQNQRGGRIFL-QDIKK-----PDRDDWENG---LTAMECALHLEKNVNQSLLELHKL
ATEK----NDPHLCDFIETHYLDEQVKAIKQLGDHVTNLRKMGAPKYG-----MAEYLF D
---KHTL-GECDN-S
>NP_001084057.1 ferritin heavy chain A [Xenopus laevis]
M-----QSQVRQNFHS-----DCEA---
-----AINRMVNMEMYASYVYLSMSYYFDRDDVALHHVAKFFKEQSHEEREHA EKFLK
YQNKRGGRVVL-QDIKK-----PERDEWGNT---LEATQAALQLEKTVNQALLDLHKL
ASDK----VDAHLCDFLESEYLEEQVKAMKQLGDYITNLKRLGVPQNG-----MGEYLF D
---KHTL-GES---S
>Anguillicola_crassus_ferritin [sequence taken from BLAST result]
-----TSQIRQNYHR-----DCEA---
-----AINRMANMELFASYTYMSMAHYFARDDVALPGFSHFFKENSEEEREHA EKLLS
FQNKRGGRIFL-NDVKK-----PERDEWGS G---LEAMQCALQLEKDVNQALLDLHKL
ASDK----IDPHLCDFLETHYLNEQVEAIKKLGDYITNLARMDSGNNK-----MAEYLF D
---KHTL-G-----
>XP_023012497.2 ferritin heavy chain [Leptinotarsa decemlineata]
MKSI----ILTTFFICVIGLASSALECSYKELDIPK-----DWIDMEK----ACVK---
-----KMRAQVEDELKAAMQYMAMGAHFSKDTVNRPGFAEIFFKSASEEREHA IKLIS
YLLMRGE--LT-SKVSSLIKRNLMPSQTTWTNG---LSALKDALKLEASVTRKIRDVIKV
CEEAK-SFNDYHLVDYLSGDFLGEQYQGQRDIAGKISTLEKMTEKHGA-----LGEWLF D
---KKLLKGEL----
>XP_040229091.2 ferritin subunit [Anopheles coluzzii]
MMKSIFFGIVALMFAAVV-MQQDQASAVTETDAPSST---DEWNYMNR----SCSA---
-----KLQDQINKEFDAAIFYMQYGAYFAQYQVNLPGFEKFFFNAASEEREHGMKLI E
YALMRGQKPID-RNSFSLNFANP-AARVDAEQGSVALTALKAAALAKEQEVTKSIRELIKI
CEEDH---NDYHLVDYLTGEFLEE QHQQRDLAGKITMLSKLLRTNPK-----LGEFMFD

---KQNM-----

>XP_049280091.1 ferritin subunit [Anopheles funestus]

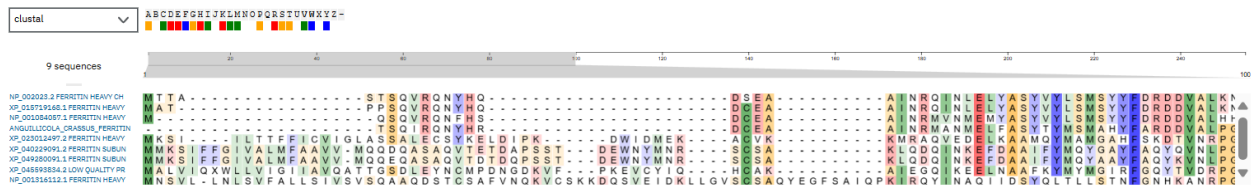
MMKSIFFGIVALMFAAVV-MQEQASAVTDTDPSSST---DEWNYMNR---SCSA---
-----KLQDQINKEFDAAIFYMQYAAFYAQYKVNLPGFENFFFKAASEEREHGMKLI
YALMRGQKPID-RNTFSLHFANQVQQPEPGPEGSVALTALKGALAKEQE-----

>XP_045593834.2 LOW QUALITY PROTEIN: ferritin heavy chain [Procambarus clarkii]

MALVIQXWLLVIGIIAVQATTGSDLEYNCMPDNGDKVF---PKEVCYIQ---HCAK---
-----AIEGQIKEELNAAFMYMGIRFGQYTVDRPGITKFLMEAATEERSHAILMLD
YLNTRGINLST-SLSYSFQF---KNYTIVYLKNMMYRDALKEALNMEIEVTKLIYDVVSA
CG-----NDYHGADVFTNPILDEQHKGVKRLQGALRAFDDLSENITGENA-VLAEIFD
---QKMLSGEFL--S

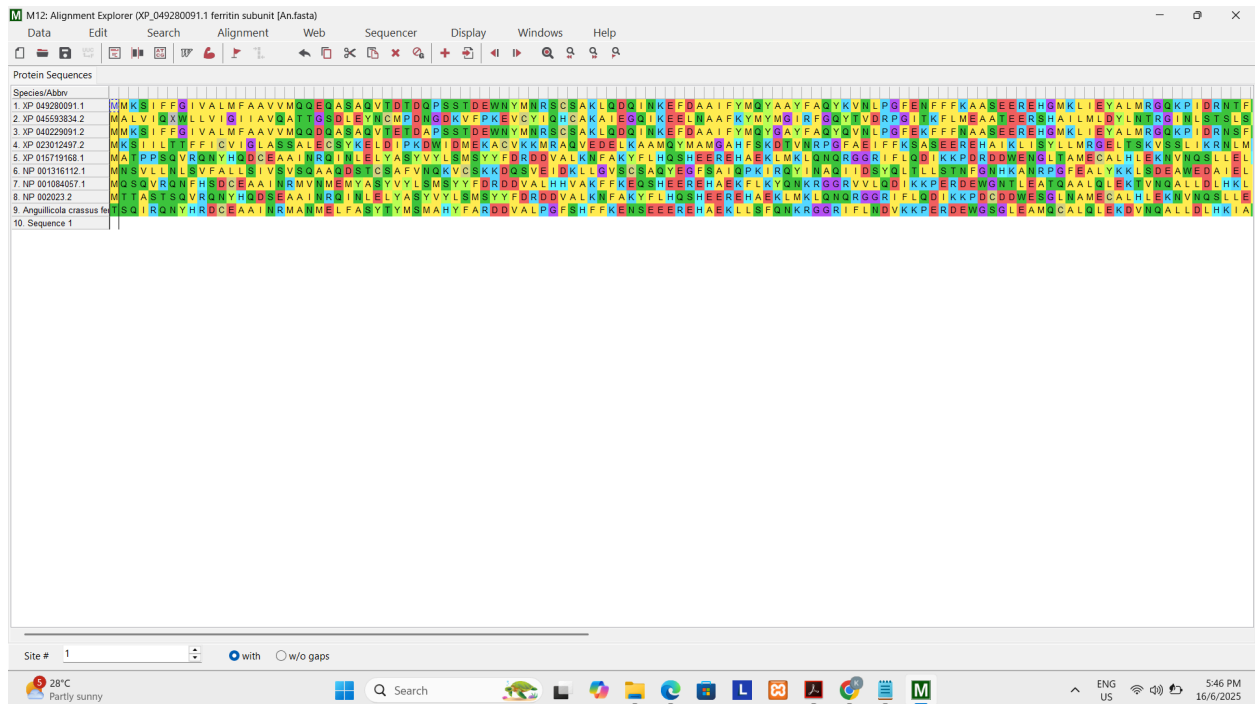
>NP_001316112.1 ferritin heavy chain-like precursor [Diaphorina citri]

MNSVL-LNLSVFALLSIVSVSQAQDSTCSAFVNQKVCCKDQSVEIDKLLGVSCSAQYE
GFSAIQPKIRQYINAQIIDSYQLTLLSTNFGNHKANRPGFEALYKKLSDEAWEDAIELIK
YNAKRGKLVDFSDIRGRDLKSMVDEQKKSAITE--YNSLATAVDLQKELAAGAHDHKE
AIRLGQAYHDPEISSFIEKEFVHKYSSNIRTLAGYLADMNSIMSEYTENGGPSLSLYLFD
EYLQKSL-GVAA--A

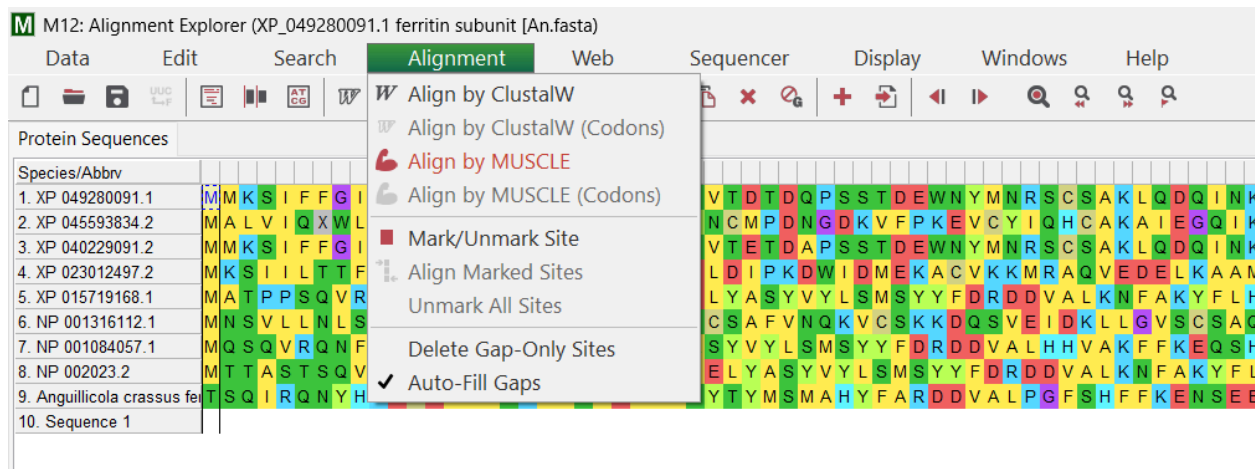


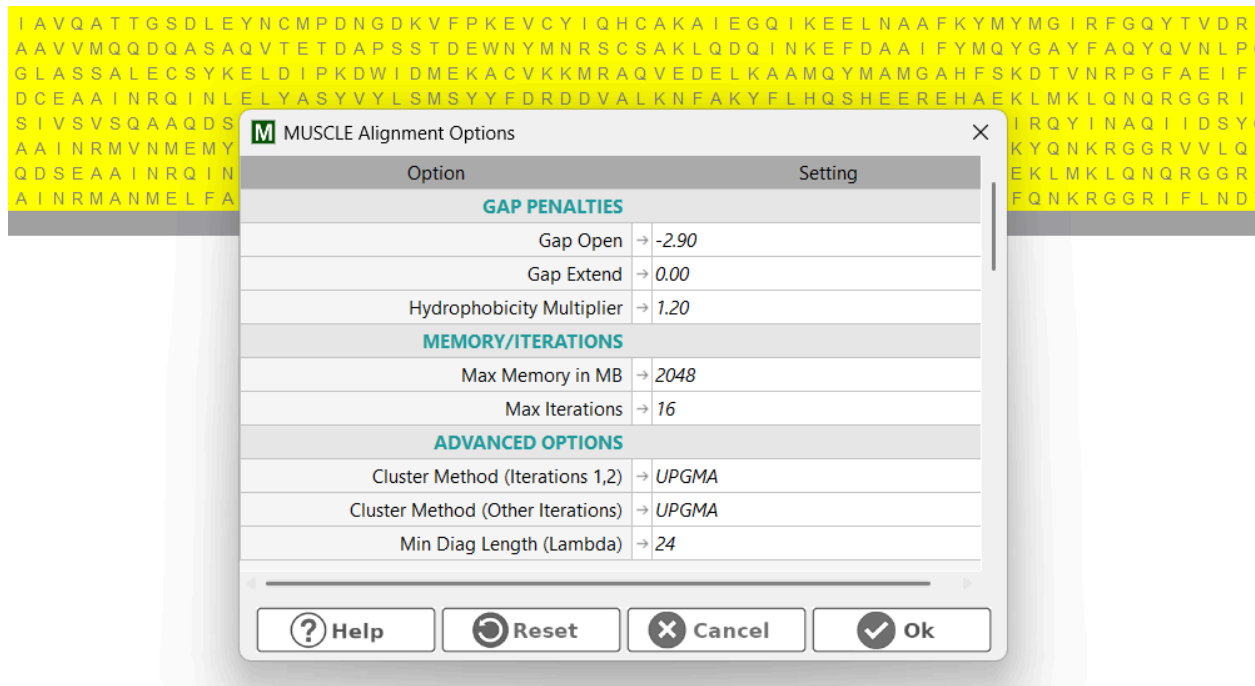
STEP 6 Phylogenetic tree construction

Using MEGA program, go Align > Create a new alignment > Protein > Edit > Insert Sequence from File > and insert the text file containing all 9 protein sequences namely the original query protein, novel protein and 8 other family proteins.



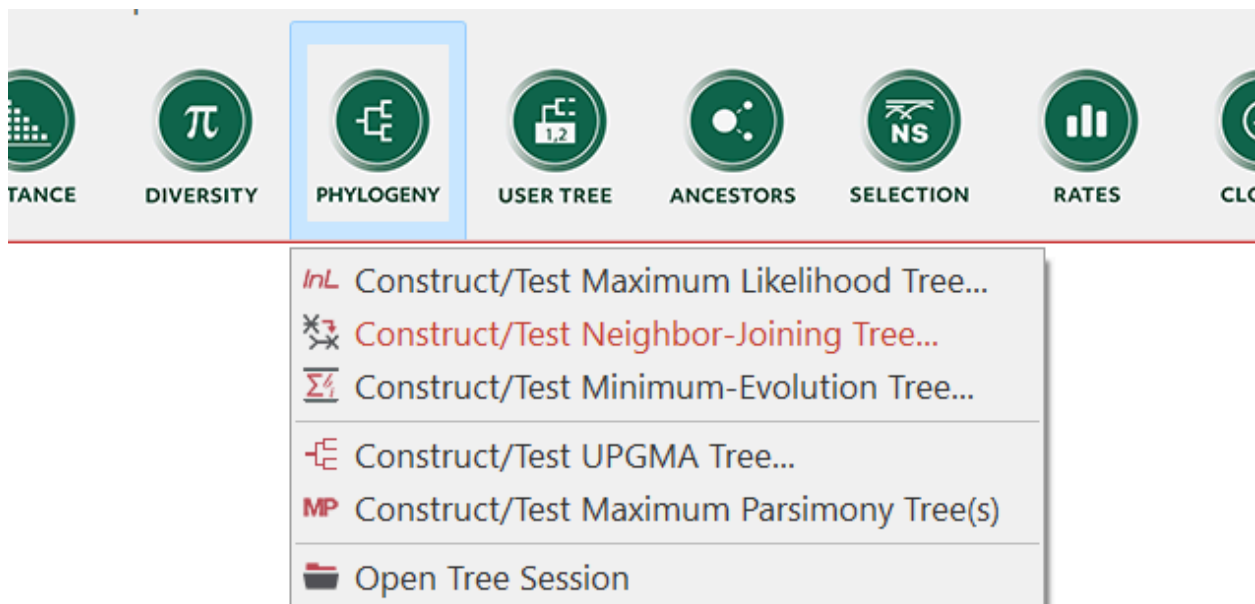
Then select all 9 sequences and click Alignment > Align by MUSCLE > the default options are kept and select Compute.





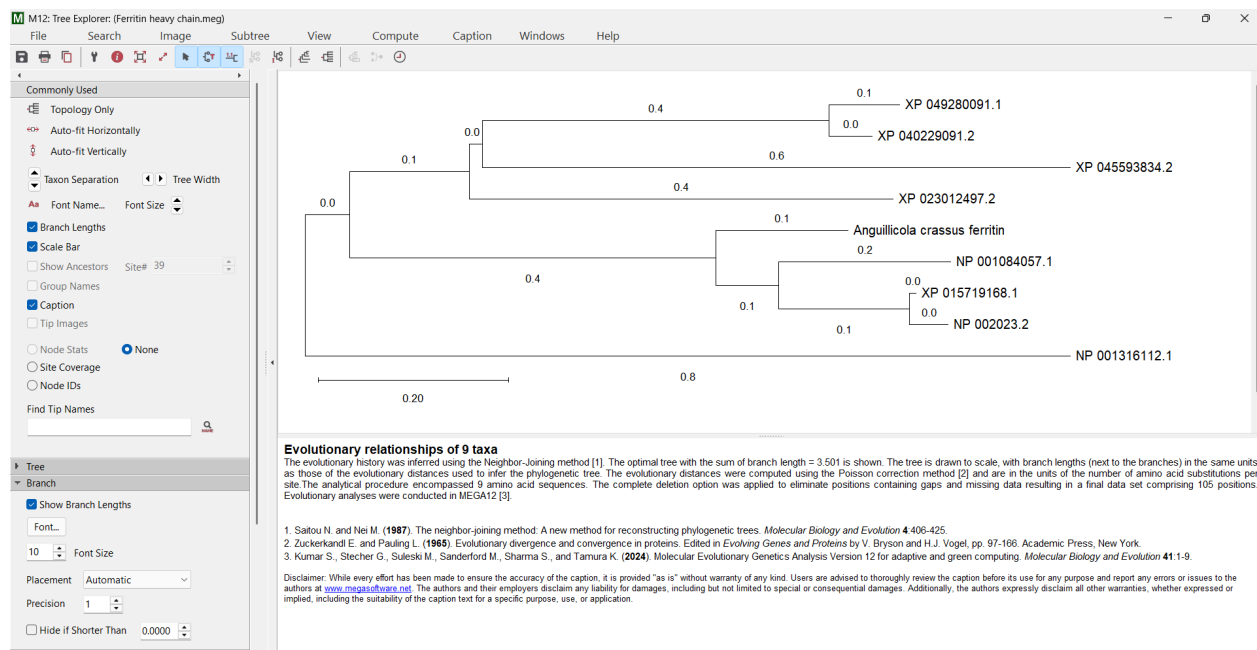
Neighbor-Joining (NJ) Tree

At the menu bar, select Phylogeny > Construct/ Test Neighbor-Joining Tree.

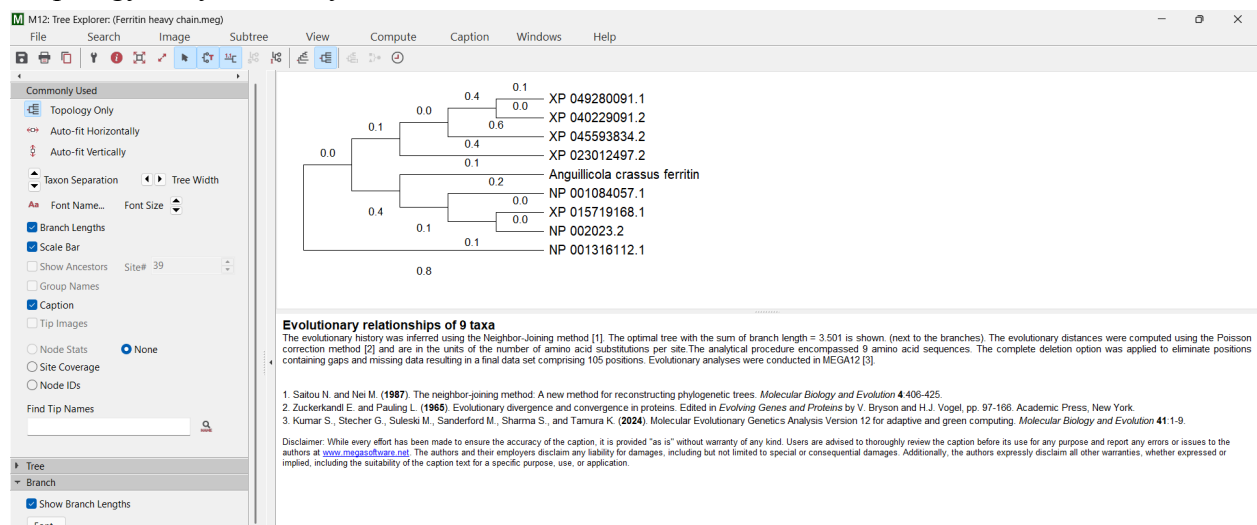


A dialog box opens with many options. Choose compute and a phylogenetic tree is produced.

Rectangular Tree Style



“Topology Only” Tree Style



Evolutionary relationships of 9 taxa

The evolutionary history was inferred using the Neighbor-Joining method [1]. The optimal tree with the sum of branch length = 3.501 is shown. The tree is drawn to scale, with branch lengths (next to the branches) in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Poisson correction method [2] and are in the units of the number of amino acid substitutions per site. The analytical procedure encompassed 9 amino acid sequences. The complete deletion option was applied to eliminate positions containing gaps and missing data resulting in a final data set comprising 105 positions. Evolutionary analyses were conducted in MEGA12 [3].

1. Saitou N. and Nei M. (1987). The neighbor-joining method: A new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution* 4:406-425.
2. Zuckerkandl E. and Pauling L. (1965). Evolutionary divergence and convergence in proteins. Edited in *Evolving Genes and Proteins* by V. Bryson and H.J. Vogel, pp. 97-166. Academic Press, New York.
3. Kumar S., Stecher G., Suleski M., Sanderford M., Sharma S., and Tamura K. (2024). Molecular Evolutionary Genetics Analysis Version 12 for adaptive and green computing. *Molecular Biology and Evolution* 41:1-9.

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Bootstrapping with Neighbor-Joining (NJ) Tree Steps similar to producing the Neighbor-Joining Tree as mentioned previously, at the dialog box, select Bootstrap method at the Test of Phylogeny and set the Number of Bootstrap Replications to 500 (default).

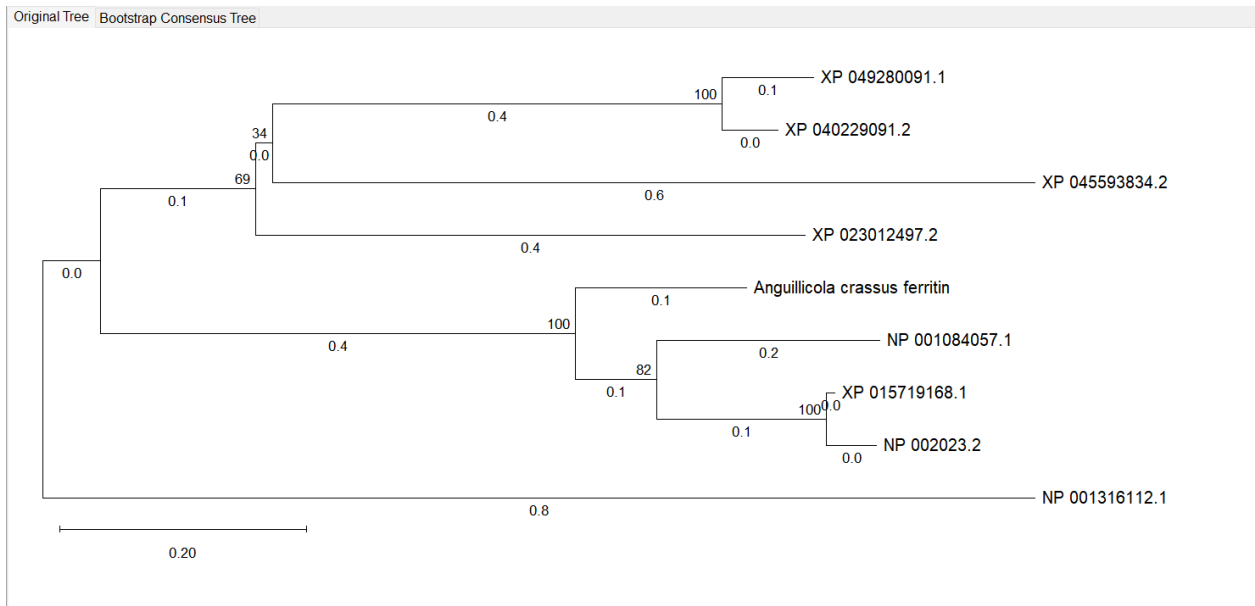
M12: Analysis Preferences

Phylogeny Reconstruction

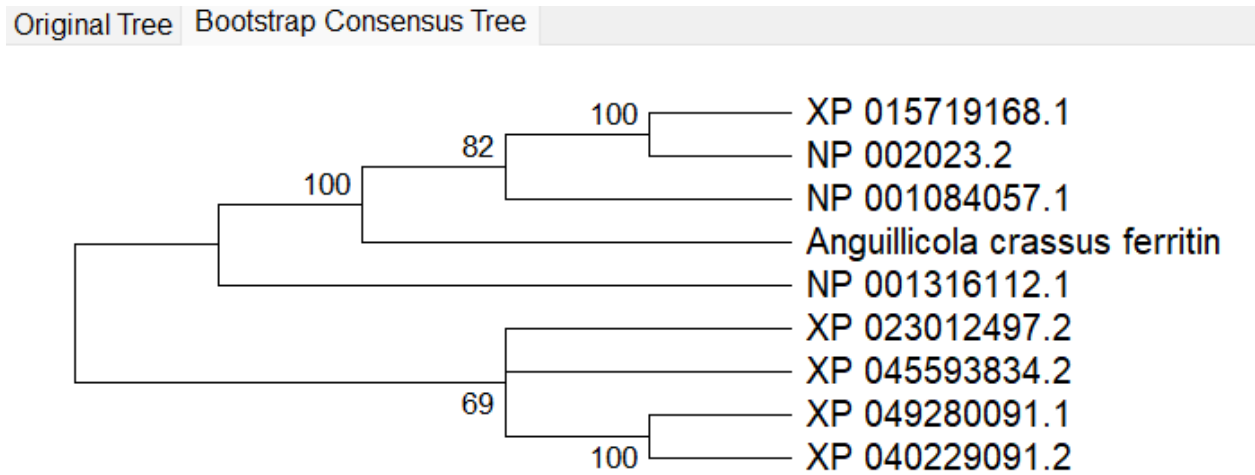
Option	Setting
ANALYSIS	
Scope	All Selected Taxa
Statistical Method	→ Neighbor-joining
PHYLOGENY TEST	
Test of Phylogeny	→ Bootstrap method
Bootstrap Replicates	→ 500
SUBSTITUTION MODEL	
Substitutions Type	→ Amino acid
Model/Method	→ Poisson model
RATES AND PATTERNS	
Rates among Sites	→ Uniform Rates
Pattern among Lineages	→ Same (Homogeneous)
DATA SUBSET TO USE	
Gaps/Missing Data	→ Complete deletion
SYSTEM RESOURCE USAGE	
Number of Threads	→ 7

? Help X Cancel ✓ Ok

Bootstrap Original Tree



Bootstrap Consensus Tree



Evolutionary relationships of 9 taxa

The evolutionary history was inferred using the Neighbor-Joining method [1]. The bootstrap consensus tree inferred from 500 replicates [2] is taken to represent the evolutionary history of the taxa analyzed [2]. Branches corresponding to partitions reproduced in less than 50% bootstrap replicates are collapsed. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (500 replicates) are shown next to the branches [2]. The evolutionary distances were computed using the Poisson correction method [3] and are in the units of the number of amino acid substitutions per site. The analytical procedure encompassed 9 amino acid sequences. The complete deletion option was applied to eliminate positions containing gaps and missing data resulting in a final data set comprising 105 positions. Evolutionary analyses were conducted in MEGA12 [4] utilizing up to 7 parallel computing threads.

1. Saitou N. and Nei M. (1987). The neighbor-joining method: A new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution* **4**:406-425.
2. Felsenstein J. (1985). Confidence limits on phylogenies: An approach using the bootstrap. *Evolution* **39**:783-791.
3. Zuckerkandl E. and Pauling L. (1965). Evolutionary divergence and convergence in proteins. Edited in *Evolving Genes and Proteins* by V. Bryson and H.J. Vogel, pp. 97-166. Academic Press, New York.
4. Kumar S., Stecher G., Suleski M., Sanderford M., Sharma S., and Tamura K. (2024). Molecular Evolutionary Genetics Analysis Version 12 for adaptive and green computing. *Molecular Biology and Evolution* **41**:1-9.

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STEP 7 Discussion

Discuss the significance of your novel gene. What have you learned about this gene/protein family?

In this *Anguillicola crassus* protein family, a species of parasitic nematode, the newly found gene is a ferritin-like protein. Ferritin is a ubiquitous iron-storage protein found in nearly all living creatures. It has a general function of keeping iron in a soluble, non-toxic, and bioavailable form, and it also plays an important function in cell defense against oxidative stress and in iron homeostasis (Arosio et al., 2009). The new gene is significant in the sense that it can reveal how parasitic nematodes manage iron metabolism in host environments, which are iron-rich and dynamic due to immune responses and host metabolic processes.

Although ferritin is well defined in vertebrates, parasitic invertebrates such as *Anguillicola crassus* have been less well investigated. This makes the new ferritin-like gene an interesting candidate for future functional studies that could provide new insights into host-parasite interaction, iron metabolism, and potentially even new targets for therapeutic intervention against parasitic infection.

Throughout this project, we were able to apply practical bioinformatics tools learned during the course of the Bioinformatics course effectively in this project. The process began with the identification of a query protein using TBLASTN searching and BLASTP analysis for novelty checking. The multiple sequence alignment, phylogenetic tree construction using Neighbor-Joining (NJ) methods, and bootstrapping tests to determine the robustness of the trees were the other activities.

Based on our findings, the Ferritin-like protein (*Anguillicola crassus*) was determined to be a new protein. This is evidenced by the fact that the percent identity of the new protein sequence and the initial *Homo sapiens* ferritin heavy chain query was not 100%, and it was also determined to have 69% identity and 85% positive matches. These are typical scores of being very similar but not close enough to be an exact homolog. Therefore, we determined that the ferritin-like gene sequence identified from *Anguillicola crassus* is new.

According to the phylogenetic trees produced, we can conclude that a phylogenetic tree's horizontal axis is employed to illustrate the degree of genetic change between sequences. Horizontal lines are branches representing evolutionary lineages over time. The longer the branch, the greater the genetic change. The bottom right scale bar provides a reference for evolutionary distance values.

Furthermore, in a bootstrapping test phylogenetic tree, red numbers at each node are bootstrap support values. Based on 500 replicates, these are the values of confidence for the

clusters of the tree. The greater the percentages, the greater the evidence that sequences emanating from a node cluster together to the exclusion of other sequences.

In short, the phylogenetic trees constructed through the Neighbor-Joining approach under the Poisson model gave varying views of evolutionary relationships amongst 9 ferritin proteins from different species. The total of the branch lengths was 4.78, which was the quantity of genetic change contained in the tree. Bootstrapping also validated the consistency of these relationships since high support values were recorded for most of the clusters, such as the clustering of *Anguillicola crassus* ferritin-like protein closely with the rest of the invertebrate ferritins. This project not only demonstrated the application of bioinformatics software in real-life scenarios, but it also contributed towards the understanding of evolutionary conservation, functional relevance, and biological importance of the ferritin protein family, particularly in parasitic organisms.